Once you have a `Paginator` instance, you access page objects inside it. Each page object indicates the items on the page and whether there are pages before or after the page. This allows you to build a template with *previous* and *next* links to continue in the pagination.

The number of the current page in a pagination sequence is a perfect use of URL query string parameters. By using a URL parameter, rather than an embedded argument, you can specify default behavior more easily.

Query parameters are accessible as part of the `HttpRequest` object that is the first argument of every view. Your request contains a special dictionary, named `GET`. This dictionary is actually an instance of a Django class called `QueryDict`, but for our purposes, you can treat it like any other dict. For any query parameters in the URL, there is a corresponding key–value pair in the `GET` dict.

To add pagination to the `musicians` view, the view needs to

- Add a `Paginator` object populated by the musicians listing
- Check for a page number query parameter
- Handle out of range page numbers
- Pass the current page to the template for rendering

The code in the following listing is the new version of the `musicians` view, applying all of these features.

Listing 4.11 Musicians listing view supporting pagination

```
# RiffMates/bands/views.py
from django.core.paginator import Paginator
...
def musicians(request):
    all_musicians = Musician.objects.all().order_by('last_name')
    paginator = Paginator(all_musicians, 2)

    page_num = request.GET.get('page', 1)
    page_num = int(page_num)

    if page_num < 1:
        page_num = 1
    elif page_num > paginator.num_pages:
        page_num = paginator.num_pages

    page = paginator.page(page_num)

    data = {
        'musicians':page.object_list,
        'page':page,
    }

    return render(request, "musicians.html", data)
```

Import the Paginator class.

Create a Paginator using the query, limiting two objects per page.

URLs are text, convert any value to an integer.

Fetch the page key from the GET dictionary, defaulting to 1 if the key does not exist.

The minimum value for the page number is 1.

The maximum value for the page number is the number of pages.

Fetch the page object containing the subset of items.

To keep the template code easy to read, pass the list of objects in as musicians.

Pass the page object in to access page number information.

The template for this view must change as well. You need to add links at the bottom for any "previous" and "next" pages. The `page` object passed into the template has